Project Report

1. Project Title

Rocket Trajectory Simulator

2. Project Objective

The purpose of this project was to better understand how launch angle and initial velocity affect a rocket's trajectory. By combining physics, mathematics, and programming, I wanted to create a simple simulation that could model rocket motion and visualize the outcome.

3. Research Question

How do launch angle and initial velocity influence a rocket's maximum height, flight time, and horizontal range?

4. Background

Rocket motion can be approximated using the principles of projectile motion when air resistance is ignored. Engineers use mathematical models to predict how different launch conditions affect flight performance.

This project explores how basic physics equations can be implemented through programming to simulate rocket trajectories and analyze flight behavior.

5. Methodology

The simulator was developed using Python.

The user provides:

- Launch Angle (degrees)
- Initial Velocity (m/s)

The program then calculates:

- Maximum Height
- Flight Time
- Horizontal Range

using standard projectile motion equations.

The calculations are based on Earth's gravitational acceleration:

$g = 9.81\ m/s^2$

6. Results

The simulation successfully calculates key flight characteristics based on user inputs.

Example:

Launch Angle = 45°

Initial Velocity = 100 m/s

Maximum Height ≈ 254.84 m

Flight Time ≈ 14.42 s

Horizontal Range ≈ 1019.37 m

The simulator demonstrates how relatively small changes in launch angle can produce significant differences in rocket performance.

7. Skills Developed

- Python Programming
- Mathematical Modeling
- Aerospace Fundamentals
- Physics-Based Simulation
- Data Analysis
- Problem Solving
- Critical Thinking

8. What I Learned

One of the most interesting things I learned from this project is that complex engineering challenges often begin with surprisingly simple physical principles.

Using Python to model projectile motion helped me better understand how mathematics can be applied to real-world aerospace problems.

This project also showed me the importance of testing, visualization, and documentation when developing engineering solutions.

9. Future Improvements

Possible future developments include:

- Air Resistance Modeling
- Earth vs Moon vs Mars Comparison
- Multi-Stage Rocket Simulation
- Trajectory Visualization Improvements
- Interactive User Interface

10. Personal Reflection

This was one of my first opportunities to combine programming, mathematics, and Aerospace Engineering concepts in a practical project.

The most rewarding part was seeing a concept I learned in physics become a working simulation that I built myself.

It reminded me that engineering is not only about learning formulas, but also about using them to better understand and solve real-world problems.

Small changes can completely change a mission.